

Notes: Milgrom and Roberts (AER, 1990)

The Economics of Modern Manufacturing: Technology, Strategy, and Organization

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1 Big picture

- **Question.** Why do the practices associated with “modern manufacturing” appear in clusters rather than as isolated changes? Why do flexible machines, fast order processing, low inventories, frequent product redesign, quality programs, design-for-manufacturability, supplier coordination, and organizational changes tend to move together?
- **Historical setting.** The paper contrasts Ford-style mass production with late-twentieth-century flexible multiproduct production. The empirical motivation is a broad set of managerial and technological changes observed in manufacturing, especially flexible machine tools, small batch sizes, just-in-time practices, rapid communication, product quality, product improvement, and team-based engineering.
- **Core idea.** Modern manufacturing practices are *complements*. The return to adopting one practice rises when the firm adopts the others. Therefore, cost declines in one technology can trigger adoption of a whole coherent strategy.
- **Methodological move.** The paper uses lattice-theoretic comparative statics and supermodularity rather than standard differentiable concave optimization. This is important because the relevant choices can be discrete, nonconvex, and mutually reinforcing.
- **Main theoretical result.** Under assumptions on demand, delay, setup/rework, and capital-cost functions, the firm’s optimized profit is supermodular in a sign-adjusted vector of strategic variables. As time passes and the cost of modern technologies falls, the optimal strategy moves upward in the modern-manufacturing direction.
- **Organizational implication.** Complementarities and nonconvexities create a need for coordination across marketing, production, order-processing, engineering, design, and supplier relations. Modern manufacturing is not just a plant-floor technology; it is a system of mutually reinforcing strategic choices.
- **Boundary result.** The number of products n is not included in the formal complementarity cluster. The model makes the ambiguous prediction that modern manufacturing can either broaden or narrow product lines, matching mixed evidence in practice.
- **Vertical organization implication.** Greater flexibility lowers the transaction-cost rationale for vertical governance of specialized assets. The modern firm may have close coordination with suppliers without formal vertical integration.

2 Incremental contribution of the paper

- **From anecdotes to a coherent economic theory.** Before this paper, many changes in manufacturing were described separately: CAD/CAM, flexible equipment, JIT, quality programs, product redesign, supplier networks, and teamwork. Milgrom and Roberts provide a unified economic logic for why these practices form a coherent cluster.
- **Complementarity as strategy design.** The paper moves complementarity from a narrow input-pair idea to a full strategy-system idea. Marketing choices, manufacturing technologies, inventory policy, quality control, engineering organization, and supplier governance can all be complementary activities.
- **Early application of supermodularity to management and industrial organization.** The paper shows how Topkis-style lattice methods can handle economic environments where differentiability, convexity, and marginal one-variable changes are not the right tools. This became influential in organizational economics and strategy.
- **Explains discontinuous adoption.** A central contribution is to show why adoption of modern manufacturing can be lumpy. If practices are complementary and the objective is nonconcave, small isolated improvements may be unprofitable, while coordinated large changes are profitable.

- **Links technology to organization.** The paper does not treat technology adoption as independent of organization. It predicts that cost reductions in flexible equipment and information processing should change coordination, supplier relations, and vertical boundaries.
- **Nuanced product-variety prediction.** The paper explicitly shows that product variety is not automatically part of the complementarity system. This is an important limitation: some modern firms broaden product lines, while others focus production.
- **Strategic-management contribution.** The paper provides a formal foundation for the idea that competitive advantage can rest on internally consistent systems of activities, not isolated best practices.

3 Data, measurement, and institutional background

3.1 No formal dataset: qualitative industrial evidence plus theory

The paper is a theoretical and interpretive essay, not an empirical paper with a formal dataset. The motivating evidence consists of managerial cases, industry reports, and examples from firms adopting flexible manufacturing, computer-aided design, just-in-time delivery, team-based engineering, and quality practices.

Interpretation: The paper’s “data” are the observed historical patterns that call for an explanation. The formal object of analysis is a model of the firm, not a regression sample.

3.2 Observed modern manufacturing cluster

The paper discusses a cluster of practices:

- flexible multipurpose equipment and programmable machines;
- smaller batch sizes and more frequent setups;
- shorter production cycles;
- lower work-in-process and finished-goods inventories;
- faster order processing and delivery;
- broader or more frequently updated product lines in some firms;
- greater emphasis on quality and fewer defects;
- design-for-manufacturability and CAD/CAM;
- cross-functional product-development teams;
- close communication with suppliers while often remaining nonintegrated.

Interpretation: The central empirical regularity is clustering. The paper wants to explain why these practices are jointly adopted and why partial adoption can be difficult or unprofitable.

3.3 Technology shocks and time parameter

The model treats time T as a proxy for exogenous technological and demand shifts. The key cost declines are:

- lower cost of data collection, organization, and communication;
- lower cost of computer-aided design and product development;
- lower cost of flexible manufacturing, setup, and retooling;

- lower cost of quality control and defect reduction;
- greater customer demand for speed and product improvement.

Interpretation: The model is comparative-static in time. As T rises, capital costs and demand conditions shift in a way that makes modern manufacturing practices more attractive.

4 Mathematical framework: complementarities and supermodularity

4.1 Lattice and sublattice ideas

A lattice is a partially ordered set in which each pair of elements has a least upper bound and greatest lower bound. In Euclidean product spaces, this means that if x and x' are feasible, then the coordinatewise maximum and minimum can also be considered.

$$\max(x, x') = (\max\{x_1, x'_1\}, \dots, \max\{x_n, x'_n\}), \quad \min(x, x') = (\min\{x_1, x'_1\}, \dots, \min\{x_n, x'_n\}). \quad (1)$$

Interpretation: This is the right language for comparing bundles of practices. A strategy is not just one real number; it is a vector of coordinated choices.

4.2 Supermodularity

A function f is supermodular if for all x, x' ,

$$f(x) + f(x') \leq f(\max(x, x')) + f(\min(x, x')). \quad (2)$$

When f is twice differentiable on a rectangle, nonnegative cross-partials imply supermodularity.

Interpretation: Supermodularity formalizes increasing differences among activities. Raising one activity raises the marginal return to raising another activity.

4.3 Monotone comparative statics

The paper uses several Topkis-style results. In words:

- the maximizers of a supermodular function over a sublattice form a sublattice;
- if a supermodular objective has increasing differences in a parameter, then the set of maximizers rises with the parameter;
- these results do not require concavity, differentiability everywhere, or continuous choices.

Interpretation: This is why the method fits manufacturing adoption. Technologies can be discrete, products can be integer-valued, and objective functions can have nonconvexities caused by complementarities.

5 Model primitives and variables

5.1 Firm decision variables

The firm is a multiproduct producer. It chooses a vector of strategic and technological variables:

- p : price of each product;
- q : expected number of improvements per product per period;
- a : order receipt and processing time;
- b : delivery time;
- c : direct marginal production cost;
- d : design cost per product improvement;
- e : extra setup cost on newly changed products;
- m : number of setups per period;
- r : probability of a defective batch;
- s : direct setup cost;
- w : wastage cost per setup;
- n : number of products, held fixed in the formal complementarity theorem.

The sign-adjusted modern-manufacturing vector is

$$(-p, m, q, -a, -b, -c, -d, -e, -r, -s, -w). \quad (3)$$

Interpretation: Higher values of this sign-adjusted vector mean lower prices, more setups, more product improvement, faster order processing and delivery, lower costs, fewer defects, and lower setup/wastage costs.

5.2 Demand, delay, and wait time

Base demand per product is

$$\mu = \mu(p, q, n, T), \quad (4)$$

and delay reduces demand through a shrinkage function

$$\delta = \delta(t, T). \quad (5)$$

Expected total delay is

$$t = a + \omega(m, r, n) + b, \quad (6)$$

where $\omega(m, r, n)$ is the expected wait for a processed order to be filled.

Realized demand is therefore

$$\mu(p, q, n, T)\delta(a + \omega(m, r, n) + b, T). \quad (7)$$

Interpretation: Speed is multidimensional: communication, order processing, production responsiveness, and delivery all matter. The convexity assumptions make reductions in different components of delay complementary.

5.3 Profit function

The total profit function is

$$\begin{aligned} \Pi(p, q, m, a, b, c, d, e, r, s, w, T) = & \left(p - c - r\rho - \frac{\tau}{m}\right) n\mu(p, q, n, T)\delta(a + \omega(m, r, n) + b, T) \\ & - m(s + w) - nq(d + e) - K(a, b, c, d, e, r, s, w, T). \end{aligned} \quad (8)$$

Here ρ is the marginal cost of reworking a defective unit, τ is inventory-holding cost per unit, and K is the capital cost of choosing technological variables.

Interpretation: The profit function links operations to marketing. More setups reduce inventory and backorders, which raises demand and permits faster response, but setups also cost money. Better design and manufacturing technology changes both direct costs and the profitability of product improvement.

6 Assumptions and theoretical results

6.1 Demand and delay assumptions A1–A4

The paper imposes assumptions on the delay function, order waiting, base demand, and capital-cost function:

- δ is decreasing and convex in delay and shifts with time;
- $\omega(m, r, n)$ is decreasing in setups m , increasing in defect probability r , and supermodular in (m, r) ;
- demand is increasing in quality/product improvement q and decreasing in price p ;
- operating profit is quasi-concave in price;
- capital-cost functions move with time as modern manufacturing and information technologies become cheaper.

Interpretation: The assumptions encode the basic economic logic: customers like speed, quality, and low price; flexible technology makes speed and redesign cheaper; defects and small batches interact through rework and waiting.

6.2 Theorem 8: coordinated rise of modern manufacturing practices

Under assumptions A1–A5 and compact feasible sets, the set of maximizers of the profit function is a compact sublattice and rises with T .

In substantive terms, as the relevant technologies become cheaper and customer demand shifts, the optimal sign-adjusted variables rise together:

- lower prices;
- lower marginal production costs;
- more frequent product redesigns and improvements;
- higher quality and fewer defects;
- faster customer communication and order processing;
- more frequent setups and smaller batches;
- lower finished-goods and work-in-process inventories;
- faster delivery;
- lower setup, wastage, and changeover costs;
- lower marginal costs of redesign.

Interpretation: The theorem formalizes the modern manufacturing cluster. A change in the cost of one technology can rationally produce coordinated changes across the entire firm strategy.

6.3 Why the adoption path can be lumpy

The model is static, but it suggests a dynamic adoption pattern:

- nonconvexities can make the optimum jump discontinuously;
- complementarity makes isolated adoption of one component relatively unattractive;
- firms may experiment initially, but once adoption is underway it should proceed with momentum;
- firms should not remain for long with flexible equipment and otherwise traditional practices.

Interpretation: This is the central organizational insight: modern manufacturing is not an incremental one-variable choice but a system transformation.

6.4 Why product-line breadth is ambiguous

The number of products n is not part of the formal complementarity cluster. The redesign cost term

$$-nq(d + e) \tag{9}$$

shows why: more products make lower redesign and changeover costs more valuable, but also make more frequent improvement q more costly.

Interpretation: The model predicts ambiguity, which matches the mixed evidence. Some flexible firms broaden product lines; others focus product lines to exploit operational benefits.

6.5 Organization and vertical governance

The paper extends the analysis to organizational variables. Let v measure the extent or complexity of vertical governance. Governance, bargaining, and distorted incentives impose costs

$$y(-v, -e), \tag{10}$$

where e indexes inflexibility or specialization of assets. Assumption A6 states that $y(-v, -e)$ is submodular. Theorem 9 then implies that as modern manufacturing flexibility rises, the optimal level of formal vertical governance falls.

Interpretation: The modern firm may need close communication with suppliers but not necessarily vertical integration. Flexible assets reduce hold-up concerns and make independent supplier relationships more attractive.

7 Figure and detailed interpretation

7.1 Figure 1: Obtaining π from Π

Read:

- Figure 1 explains why the authors transform the profit function before applying supermodular comparative statics.
- The left panel shows the underlying operating-profit function as a function of price.
- The maximized profit over price need not inherit the right monotonicity properties if price is left as a direct decision variable.

- The solution is to replace price with a lower bound $\bar{p}$ and define the optimized function π over the remaining variables and $\bar{p}$.
- The sign convention then treats lower price as part of the modern manufacturing cluster.

Interpretation: The figure is technical but important. It shows how the authors avoid a problem created by price optimization. They want a supermodular objective in sign-adjusted choices, and optimizing over price directly can break that structure. Introducing a lower bound allows the model to include lower prices in the complementarity system.

Economic message: Modern manufacturing makes lower prices compatible with quality and speed because lower costs, faster setups, and lower inventories raise the return to aggressive pricing. Figure 1 helps incorporate that idea into the supermodular model.

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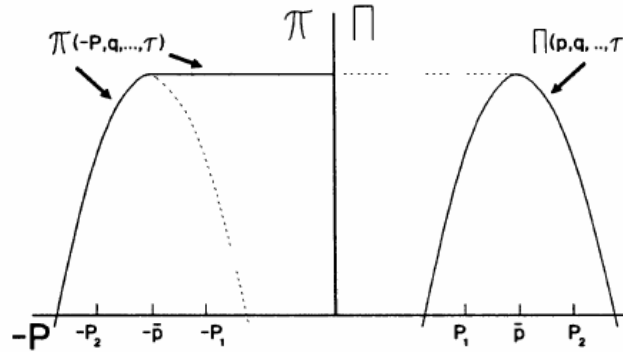


FIGURE 1. OBTAINING π FROM Π

NOTE THAT

Figure 1: Obtaining π from Π .

8 Theoretical identification strategy

8.1 Step 1: Identify a clustered empirical phenomenon

The paper begins from a stylized fact: modern manufacturing practices appear together. The first identification move is qualitative: identify a broad cluster of practices across technology, marketing, production, engineering, and organization.

8.2 Step 2: Reinterpret the cluster as complements

The paper then translates the cluster into a sign-adjusted vector of strategic variables and shows that many cross-effects are nonnegative. This identifies the common economic force behind co-adoption: increasing differences.

8.3 Step 3: Use time as a technology-cost shifter

The model treats the passage of time as exogenous shifts in the costs of information processing, flexible equipment, CAD, and redesign. These shifts raise the optimal level of the whole complementary strategy cluster.

8.4 Step 4: Explain adoption discontinuities through nonconvexities

The theory does not predict smooth adjustment by every firm. It predicts clusters and potential jumps. Empirically, support would come from observing firms separated into coherent systems rather than continuous mixtures of independent practices.

8.5 Step 5: Connect technology to organization

The same complementarity logic extends to coordination and vertical governance. Integrated planning across functions becomes more important, while formal vertical ownership of specialized assets can become less important as assets become more flexible.

9 Interpretive synthesis

9.1 Why the paper matters for industrial organization

The paper uses price theory to explain strategy systems. It shows that technological change can reorganize firms not only by changing production costs but by changing the returns to a whole vector of practices.

9.2 Why the paper matters for management

The theory gives an economic foundation for “fit” among activities. A firm cannot evaluate CAD, JIT, quality systems, product redesign, and supplier networks independently because the payoff to one depends on the others.

9.3 Why supermodularity is essential

Standard concave calculus is designed for smooth marginal adjustments. Modern manufacturing adoption is often discrete and system-wide. Supermodularity handles this by studying the order structure of choices rather than relying on local first-order conditions.

9.4 What the paper does not do

The paper does not provide a formal empirical test. It also abstracts from competition among firms, learning costs, financing constraints, and heterogeneity across industries. Its contribution is a general framework rather than a calibrated or estimated model.

10 Short summary

- The paper explains why modern manufacturing practices cluster together.
- The practices include flexible equipment, small batches, low inventories, quality, speed, product redesign, CAD/CAM, supplier coordination, and organizational redesign.

- The central concept is complementarity: adopting one practice raises the return to adopting others.
- The mathematical tool is supermodularity and lattice-theoretic comparative statics.
- The model includes eleven sign-adjusted decision variables and a time parameter.
- Under the assumptions, optimal modern-manufacturing practices rise together as technology and demand change.
- Nonconvexities imply lumpy adoption and coordinated transformation.
- Product-line breadth is not unambiguously part of the complementarity cluster.
- Modern manufacturing requires coordination across firm functions.
- Greater asset flexibility can reduce the need for formal vertical integration even while supplier communication rises.

11 Conclusion

11.1 Core conclusion

Milgrom and Roberts argue that modern manufacturing is a coherent system of complementary practices. Flexible technology, speed, quality, low inventories, frequent product redesign, CAD/CAM, and organizational coordination are not independent improvements; they are mutually reinforcing elements of a profit-maximizing strategy.

11.2 Economic intuition

When setup costs fall, smaller batches become attractive. Smaller batches reduce inventories and make faster order processing valuable. Faster response supports lower prices and higher quality. CAD and design-for-manufacturability lower redesign costs and make flexible equipment more valuable. Each improvement raises the return to the others.

11.3 Incremental contribution in one sentence

The paper's lasting contribution is to formalize the idea of strategic fit using supermodularity, showing why technological change can induce coordinated, discontinuous changes in manufacturing, marketing, engineering, and organizational design.

11.4 Takeaway

The modern manufacturing firm should be understood as a system. Firms that imitate only one visible practice may fail because the value of that practice depends on complementary changes elsewhere in the organization.